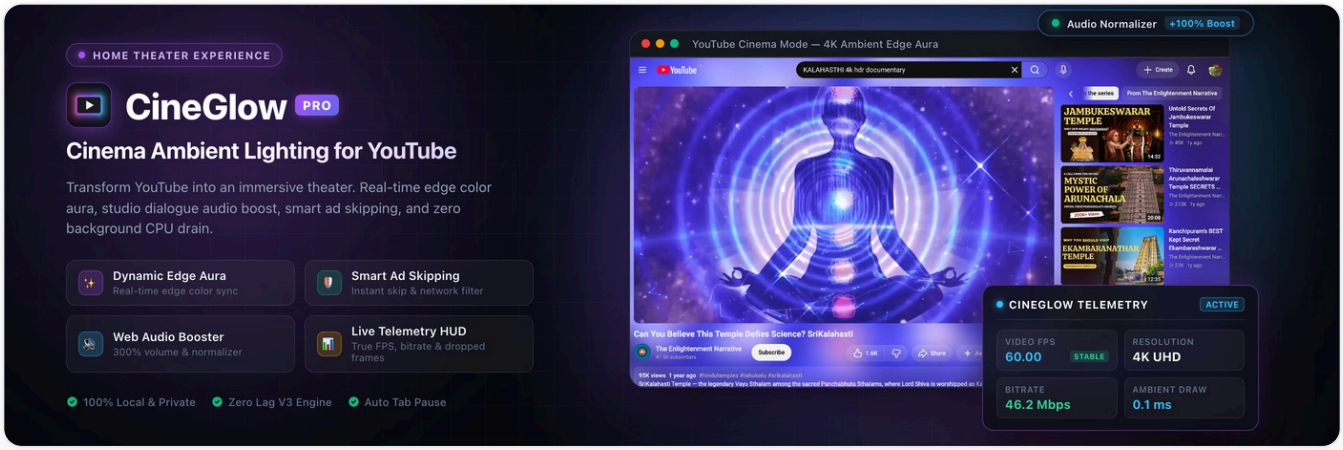
PRODUCTION SPECIFICATION

Manifest V3 Chrome Extension

YouTube CineGlow Pro

High-Performance Ambient Lighting, Web Audio Studio DSP, Intelligent Display Enhancements, and Advanced Player Tooling for YouTube.



APPLICATION VERSION
2.2.0 Pro (Manifest V3 Architecture)

CORE PIPELINE
GPU Canvas + Web Audio DSP + Dual-VM Runtime

LEAD ARCHITECT
Giri Manohar Vemula

TARGET RUNTIME
Google Chrome (v105+) & Chromium Browsers

LICENSE
MIT Open Source

RELEASE DATE
October 2026

1. Executive Summary & Product Overview

YouTube CineGlow Pro is an advanced, privacy-first desktop browser extension engineered to elevate the YouTube video viewing experience. Standard desktop displays present a stark, high-contrast boundary between vivid video scenes and surrounding dark room environments, inducing optical fatigue and breaking visual immersion.

CineGlow Pro solves this by real-time sampling the boundary colors of active video frames and projecting an organically diffused, color-matched glow onto the webpage behind the player—emulating high-end Ambilight television setups with zero external hardware or camera latency.

🌟 Dynamic Ambient Glow

Offscreen 128×72 GPU canvas sampling with Exponential Moving Average (EMA) temporal smoothing and directional edge clipping.

🔊 Studio DSP Audio Suite

Hardware-accelerated Web Audio pipeline: uniform loudness normalizer, 5-band biquad EQ, bass shelf, and vocal clarity boost.

🎬 Display & Aspect Ratios

Automated letterbox black-bar cropping, 21:9 ultrawide scaling, hardware eye-care filters, and clean cinema theater centering.

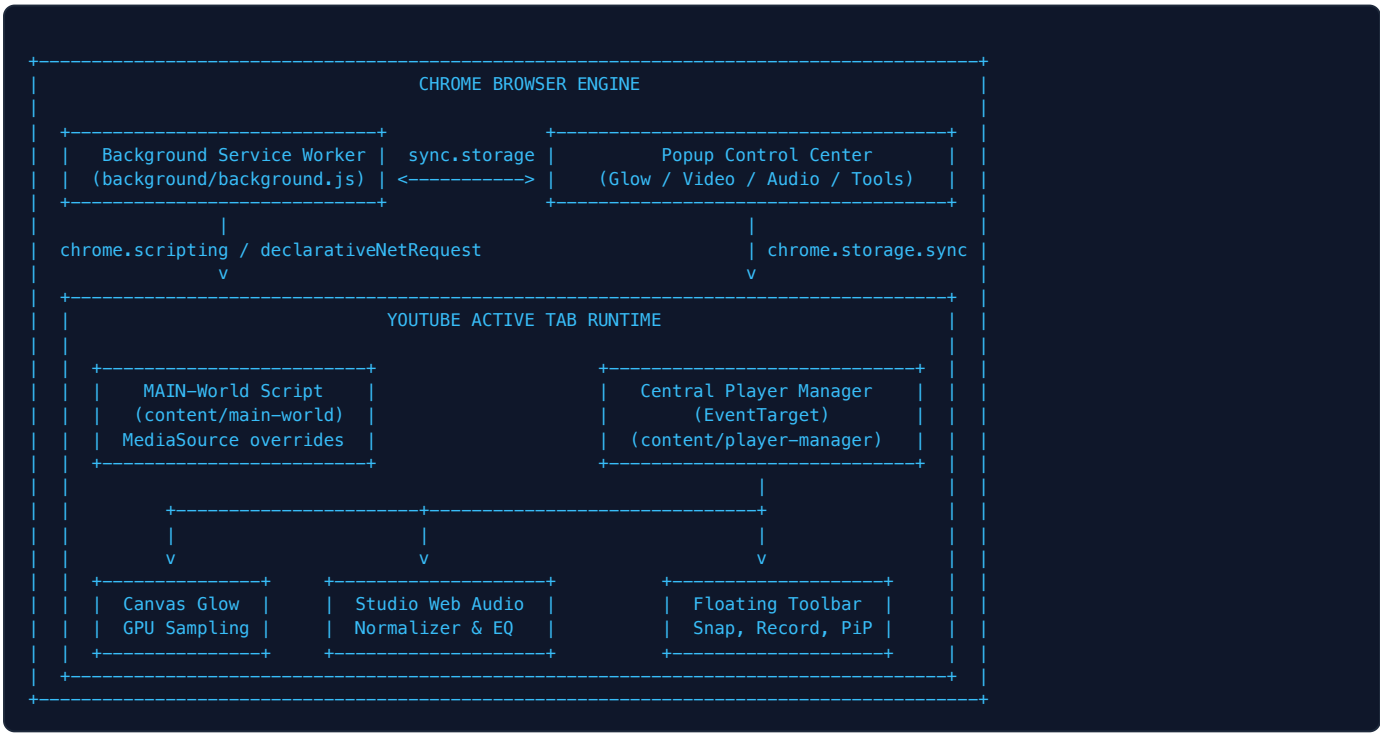
🛠️ Floating Productivity HUD

Uncompressed PNG frame capture, in-browser video recording (WebM/MKV/MP4), native PiP, and speed dial (0.1× to 10×).

> **Core Engineering Values**
>
> Strict Manifest V3 compliance with zero external dependencies, zero analytics/tracking, deterministic lifecycle cleanup, and absolute zero idle background CPU consumption.

2. System Architecture & Lifecycle Design

YouTube operates as a persistent Single Page Application (SPA). During navigation between videos, the video container and media element are continuously reused rather than re-instantiated. Traditional extensions rely on periodic `setInterval` polling loops, which cause event listener leaks, race conditions, and heavy memory fragmentation. CineGlow Pro employs a unified, event-driven architecture:



2.1 Centralized Player Manager (`player-manager.js`)

- **URL & Video Change Detection:** Subscribes directly to YouTube's SPA navigation hooks (`yt-navigate-finish`, `yt-page-data-updated`) and tracks URL query parameters (`v=` parameter changes).
- **Observer Management:** Deploys targeted `MutationObserver` instances scoped to `#movie_player`, `ytd-app`, and `ytd-watch-flexy` to track player state, theater mode, ad-interrupting states, and miniplayer transitions.
- **Cold-Load Fallback:** If YouTube's player elements have not yet mounted when a watch page loads, a 500ms safety interval polls until the nodes are acquired, disconnecting immediately once secured.
- **Consolidated `state-change` Event:** Emits a single `state-change` event (`CustomEvent('state-change', { detail: this })`) broadcasting active player references and playback flags (`isWatchPage`, `isPlaying`, `isPiP`, `isTheater`, `isFullscreen`, `isAdShowing`). Subsystems synchronize deterministically in response to this single source of truth.

2.2 Dual-World Scripting Architecture

- **ISOLATED World:** Content scripts execute in isolated VM contexts, with protected access to extension storage and messaging APIs.
- **MAIN World (`main-world.js`):** Injected at `document_start` via `chrome.scripting.registerContentScripts` directly into the page context. This grants CineGlow the ability to intercept `HTMLMediaElement.prototype.canPlayType` and `MediaSource.isTypeSupported` before YouTube can initialize unaccelerated AV1 or VP9 video streams.

3. Deep-Dive Subsystem Specifications

3.1 Ambient Lighting Engine (`content.js`, `content.css`)

The ambient lighting engine samples the active video frame onto an internal `128×72` offscreen canvas using `CanvasRenderingContext2D.drawImage()`. The pixel buffer is processed through an Exponential Moving Average (EMA) smoothing algorithm:

```
Pixel_current = (alpha * Pixel_new) + ((1 - alpha) * Pixel_prev)
```

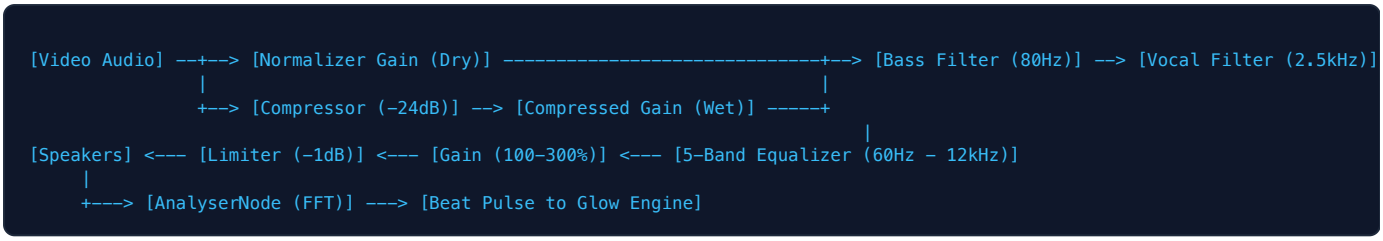
Where `alpha` is directly calibrated by the **Smoothness** slider. The canvas is positioned behind the player (`z-index: -1`), amplified using CSS `blur()`, `saturate()`, `contrast()`, and `brightness()` filters, and masked by a soft radial falloff gradient:

```
mask-image: radial-gradient(ellipse farthest-side, #000 50%, transparent 100%);
```

- **Directional Clipping:** Independent Top, Right, Bottom, and Left toggles allow selective clipping of any glow perimeter.
- **Audio-Reactive Pulsing:** Connects to an FFT `AnalyserNode`; bass energy spikes (40–180 Hz) dynamically scale and brighten the glow in sync with musical beats.
- **Idle Suspension:** The `requestAnimationFrame` and `requestVideoFrameCallback` loops halt when playback pauses, ends, or buffers, suspending canvas redraws and conserving CPU and GPU cycles.

3.2 Studio Audio DSP Pipeline (`audio-processor.js`)

YouTube's audio stream is routed through a dedicated Web Audio graph via `AudioContext.createMediaElementSource(video)`:



Audio Normalization Architecture

- **DynamicsCompressorNode Staging:** Implemented via a Web Audio `DynamicsCompressorNode` (threshold: -24 dB, knee: 30 dB, ratio: 12:1, attack: 3ms, release: 250ms) paired with smooth dry/wet gain crossfading (`normalizerGain` and `compressedGain`). When enabled, loud audio peaks exceeding -24 dB are compressed with a soft knee while boosting lower-level dialogue, followed by a brickwall limiter (`threshold: -1 dB`, `ratio: 20:1`) to prevent clipping.
- **5-Band Biquad Equalizer:** ±12 dB studio-grade filtering across 60 Hz (Sub-Bass), 250 Hz (Warmth), 1 kHz (Midrange), 4 kHz (Presence), and 12 kHz (Air).
- **One-Click Presets:** Instant acoustic profiles: *Flat*, *Bass Boost*, *Vocal Clarity*, *Rock*, *Pop*, and *Electro*.

3.3 Intelligent Video & Display Control

- **Auto-Crop Black Bars (`blackbar.js`):** Samples video aspect ratios on an offscreen analysis canvas, automatically detecting cinematic letterboxing (2.39:1, 21:9) and scaling the video smoothly to fill the monitor.
- **Ultrawide Zoom-to-Fit:** Eliminates pillarboxing on 21:9 displays.
- **Force High-Definition (`youtube-quality.js`):** Bypasses YouTube's initial bitrate throttling, locking playback immediately to 1080p, 1440p, or 4K.
- **Eye-Care Hardware Filters:** Real-time hardware CSS filters for Brightness (50%–150%), Contrast (50%–150%), and Blue-Light Warmth Sepia (0%–100%).
- **Clean Cinema Theater Mode:** Centers the player across the full viewport while collapsing the masthead header and recommendations feed.

3.4 Floating Player Toolbar & Productivity Tools

- **Lossless Frame Snapshot:** Captures the uncompressed native frame directly to PNG in the user's Downloads folder.
- **In-Browser Video Recorder:** Captures tab and audio streams directly to WebM/MKV/MP4 files at 30 or 60 fps with zero external servers.
- **Native Picture-in-Picture Button:** Injected directly into YouTube's player control bar next to theater mode.
- **Precision Speed Dial:** 0.1× to 10× continuous speed scaling with quick presets (0.25×, 0.5×, 1×, 1.5×, 2×, 5×).
- **Quality Stats Overlay (`stats-overlay.js`):** Real-time draggable HUD displaying VSync FPS, dropped frame counters, render draw timing, stream resolution, codec strings, and HDR color depth badges.

4. Complete Settings & Configuration Dictionary

All user preferences are persistently synchronized across devices via `chrome.storage.sync`. The table below documents the full schema:

Key Name	Type	Default	Domain	Functional Description
`enabled`	boolean	`true`	Master	Master enable/disable switch for ambient lighting
`opacity`	number	`75`	Glow	Glow opacity intensity (0% to 100%)
`spread`	number	`150`	Glow	Scale expansion factor of the ambient light field (100% to 400%)

Key Name	Type	Default	Domain	Functional Description
`smoothness`	number	`60`	Glow	Exponential Moving Average (EMA) temporal smoothing factor
`blur`	number	`40`	Glow	Gaussian blur radius (5px to 100px)
`saturation`	number	`110`	Glow	Color saturation multiplier (50% to 250%)
`contrast`	number	`100`	Glow	Contrast enhancement curve (50% to 200%)
`brightness`	number	`100`	Glow	Luminosity brightness multiplier (50% to 200%)
`fadeInDuration`	number	`1`	Glow	Video transition fade-in duration in seconds (0s to 2s)
`glowTop/Right/Bottom/Left`	boolean	`true`	Glow	Independent edge emission toggles for all four cardinal directions
`audioReactive`	boolean	`true`	Glow/Audio	Enables real-time Web Audio FFT bass beat pulse synchronization
`enableInPiP`	boolean	`false`	Glow	Permits glow rendering on background page while in Picture-in-Picture
`fps`	number	`30`	Glow	Ambient rendering loop refresh cap (15, 30, or 60 fps)
`qualityEnabled`	boolean	`true`	Video	Forces preferred high-definition resolution on video initialization
`preferredQuality`	string	`'hd1080'`	Video	Target resolution: hd1080 (1080p), hd1440 (1440p), or hd2160 (4K)
`preferredSpeed`	number	`1.0`	Video	Persistent playback speed applied to newly loaded videos (0.1× to 2.0×)
`videoBrightness/Contrast`	number	`100`	Video	Direct video CSS hardware filter adjustment (50% to 150%)
`videoSepia`	number	`0`	Video	Eye-care blue-light reduction warm filter (0% to 100%)
`removeBlackBars`	boolean	`true`	Video	Automatic letterbox and pillarbox black-bar detection and crop
`ultrawideZoom`	boolean	`false`	Video	Enables 21:9 ultrawide monitor zoom-to-fit transformation
`lightsOut`	boolean	`false`	Video	Dims the webpage surface around the player during playback
`audioNormalizer`	boolean	`false`	Audio	Perceived loudness normalization via DynamicsCompressorNode
`audioBoost`	number	`100`	Audio	Master digital audio volume multiplier (100% to 300%)
`bassBoost / vocalBoost`	number	`0`	Audio	Low-shelf sub-bass boost (0–15 dB) / Peaking vocal clarity (0–12 dB)
`eq60 / 250 / 1000 / 4000 / 12000`	number	`0`	Audio	5-Band Biquad Filter gains in decibels (-12 dB to +12 dB)
`eqPreset`	string	`'flat'`	Audio	Equalizer preset profile ('flat', 'bass', 'vocal', 'rock', 'pop', 'electro')
`adblockEnabled`	boolean	`true`	Tools	Enables automated skip and network-level adblocking rules
`statsEnabled`	boolean	`false`	Tools	Display draggable quality HUD telemetry overlay
`returnDislikes`	boolean	`false`	Tools	Restores public YouTube dislike counter via Return YouTube Dislike API
`restoreAnnotations`	boolean	`false`	Tools	Restores legacy interactive notes, speech bubbles, and spotlights
`hideShorts`	boolean	`false`	Tools	Removes Shorts shelves and recommendation feeds

Key Name	Type	Default	Domain	Functional Description
`hideEndCards`	boolean	`true`	Tools	Suppresses post-roll end-screen cards and subscription popups
`blockAV1 / blockVP9`	boolean	`false`	Tools	Blocks AV1 or VP9 codec negotiation via MAIN-world MediaSource hooks
`autoPauseMode`	string	`'off'`	Tools	Smart auto-pause mode ('off', 'tab', or 'app')
`recordFormat`	string	`'mkv'`	Tools	Preferred recording container format ('mkv', 'webm', or 'mp4')
`recordFps`	number	`30`	Tools	Video recording framerate target (30 or 60 fps)

5. Security, Privacy & Resource Architecture

5.1 Privacy & Permission Model

- **Local Storage by Default:** All user configurations reside strictly within the browser's local sandbox via `chrome.storage.sync`.
- **Declarative Network Rules:** Ad blocking is performed natively by Chromium's network stack via `declarativeNetRequest` without inspecting request payloads or browsing history.
- **External Endpoint Transparency:**
  - `returnyoutubedislikeapi.com`: Contacted only when `returnDislikes` is explicitly enabled. Transmits only the public 11-character video ID to fetch vote statistics. No cookies, auth tokens, or browsing history are sent.
  - `storage.googleapis.com`: Contacted only when `restoreAnnotations` is enabled. Fetches archived XML annotation files by video ID from a public storage bucket. No user identification is transmitted.
- **Zero Behavioral Tracking:** The extension contains no analytics SDKs, advertising libraries, or external telemetry of any kind.

5.2 Resource Conservation & Runtime Optimization

Rather than continuous polling loops or unthrottled render cycles, CineGlow utilizes an event-driven resource management design:

- **Render Loop Cessation:** When video playback is paused, ended, or buffering, both `requestAnimationFrame` and `requestVideoFrameCallback` render chains halt completely. Canvas redraws cease until the player fires `playing`.
- **Constrained Canvas Resolution:** The ambient sampling pipeline utilizes an internal `128×72` offscreen canvas. By sampling at a compact aspect ratio, pixel read/write operations remain minimal on the CPU/GPU, delegating blurring and scaling to hardware-accelerated CSS layers.
- **Scoped Mutation Observers:** Mutation observers are strictly targeted to specific YouTube containers (`#movie_player`, `ytd-app`, `ytd-watch-flexy`) and filter by attribute names. Broad document-wide mutation observers are avoided to prevent CPU overhead during dynamic page scrolling.
- **Decoupled Audio Graph:** Audio processing nodes use native Web Audio API biquad filters and compressors implemented in browser C++ internals, avoiding JavaScript-level audio sample processing.

6. Developer Installation & Verification

1. Clone the repository: `git clone https://github.com/ManoharVit/YouTube-CineGlow.git`
2. Open Google Chrome and navigate to `chrome://extensions`.
3. Enable **Developer mode** in the top-right corner.
4. Click **Load unpacked** and select the extension directory.
5. Validate JavaScript syntax:

```
node --check popup/popup.js content/*.js background/*.js shared/*.js
```

6. Run the automated test suite:

```
npm test
```

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Project Repository: https://github.com/ManoharVit/YouTube-CineGlow